

# TestPlatform

## Rapport d'Execution de Campagne

**test**

Projet: ms\_gestion

Version: 2.5

Environnement: ms\_gestion\_env

Debut: 29/06/2026 15:07 | Fin: 29/06/2026 15:08

Duree: 88.9 s

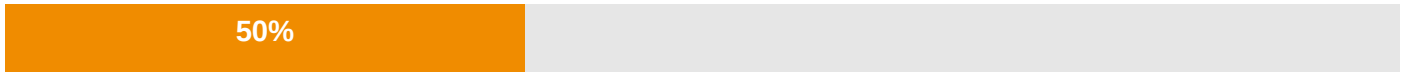
Rapport genere le 29/06/2026 15:08

**PARTIEL**

## Resume executif

| Total | Succes | Echecs | Erreurs | Duree  |
|-------|--------|--------|---------|--------|
| 2     | 1      | 1      | 0       | 88.9 s |

### Taux de reussite



Evolution vs campagne precedente: N/A

Premiere campagne

**Corrections IA: 2 test(s) corrige(s) automatiquement, 1 avec succes**

## Informations du contexte

|             |            |
|-------------|------------|
| Projet      | ms_gestion |
| Description | test       |

## Suites

| Nom               | Type        | Depot Git | Branche |
|-------------------|-------------|-----------|---------|
| Suite UNIT        | UNIT        | N/A       | main    |
| Suite INTEGRATION | INTEGRATION | N/A       | main    |

## Cas de test

| Titre              | Type        | Priorite | Risque   | Script |
|--------------------|-------------|----------|----------|--------|
| create project     | UNIT        | 2        | HIGH     | IA     |
| create envirenemnt | INTEGRATION | 5        | CRITICAL | IA     |

## Environnement

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| URL Web         | http://localhost:3000                                                              |
| URL API         | N/A                                                                                |
| Repository      | https://github.com/ilyeschaabani/Automatisation_intelligente_des_tests_applicatifs |
| Branche         | main                                                                               |
| Base de donnees | POSTGRESQL                                                                         |

# Resultats details

## Synthese par suite

| Suite             | Total | Succes | Echecs | Taux |
|-------------------|-------|--------|--------|------|
| Suite UNIT        | 1     | 1      | 0      | 100% |
| Suite INTEGRATION | 1     | 0      | 1      | 0%   |

## Suite UNIT

create project

IA x1

SUCCESS

|        |      |        |                       |
|--------|------|--------|-----------------------|
| Type   | UNIT | Duree  | 11.5 s                |
| Script | IA   | Erreur | [COMPILATION_ERROR] — |

## Suite INTEGRATION

create envirenemnt

IA x1

FAILURE

|        |             |        |                                                                                                                                            |
|--------|-------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Type   | INTEGRATION | Duree  | 9.3 s                                                                                                                                      |
| Script | IA          | Erreur | [COMPILATION_ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project ms_gestion: |

# Analyse IA

## create project

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Erreurs<br>extraites | <pre>[INFO] Scanning for projects... [INFO] [INFO] -----&lt; com.pfe.platform:ms_gestion &gt;----- [INFO] Building ms_gestion 0.0.1-SNAPSHOT [INFO]    from pom.xml [INFO] -----[ jar ]----- [INFO] [INFO] --- resources:3.3.1:resources (default-resources) @ ms_gestion --- [INFO] Copying 1 resource from src\main\resources to target\classes [INFO] Copying 0 resource from src\main\resources to target\classes [INFO] [INFO] --- compiler:3.14.1:compile (default-compile) @ ms_gestion --- [INFO] Nothing to compile - all classes are up to date. [INFO] [INFO] --- resources:3.3.1:testResources (default-testResources) @ ms_gestion --- [INFO] skip non existing resourceDirectory C:\tmp\ms-execution\exec- 3421550725211656858\Backend\ms_gestion\src\test\resources [INFO] [INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ ms_gestion --- [INFO] Recompiling the module because of changed source code. [INFO] Compiling 2 source files with javac [debug parameters release 17] to target\test-classes [INFO] [INFO] --- surefire:3.2.5:test (default-test) @ ms_gestion --- [INFO] Using configured provider org.apache.maven.surefire.testng.TestNGProvider [INFO] [INFO] ----- [INFO]  T E S T S [INFO] ----- [INFO] Running suites.unit.ProjectServiceCreateHappyTest [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.127 s -- in suites.unit.ProjectServiceCreateHappyTest [INFO] [INFO] Results: [INFO] [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0 [INFO] [INFO] ----- [INFO] BUILD SUCCESS [INFO] ----- [INFO] Total time:  7.497 s [INFO] Finished at: 2026-06-29T14:35:36+01:00 [INFO] -----  ----- STDERR ----- Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended</pre> |
| Analyse<br>IA        | <pre>### 1. Ce qui est bien testé Le test couvre la logique *happy path* de la création de projet : validation de l'unicité du nom, sauvegarde du projet, création de l'enregistrement utilisateur comme membre, et retour des bonnes valeurs dans la réponse. L'utilisation de `MockedStatic` pour `SecurityUtils.getCurrentUserId()` et la simulation correcte des dépendances (`projectRepository`, `projectMemberRepository`) garantissent un contrôle précis de l'environnement de test.  ### 2. Suggestion d'amélioration</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Le test manque de **\*\*vérification de l'ordre ou de la cohérence entre les appels\*\*** (ex. : garantir que le projet est sauvegardé **\*avant\*** la création du membre). De plus, les assertions sur la réponse pourraient inclure ``assertEquals(result.getId(), 1L)`` pour valider que l'ID est bien renseigné, améliorant ainsi la robustesse. Un petit bémol : on pourrait extraire la configuration des mocks (``when(...).thenReturn(...)``) dans une méthode helper pour une lisibilité accrue.

## create envirenemnt

```
Erreurs [ERROR]
extraites [INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default
ms_gestion:
[ERROR]
[ERROR] Please refer to C:\tmp\ms-execution\exec-12815052988845666633\Backend\ms_gestion\target
the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date]
[ERROR] There was an error in the forked process
[ERROR]
[ERROR] Cannot instantiate class suites.integration.EnvironmentServiceIT
[ERROR] org.apache.maven.surefire.booter.SurefireBooterForkException: There was an error in the
[ERROR]
[ERROR] Cannot instantiate class suites.integration.EnvironmentServiceIT
[ERROR]   at org.apache.maven.plugin.surefire.booterclient.ForkStarter.fork(ForkStarter.java:
[ERROR]   at org.apache.maven.plugin.surefire.booterclient.ForkStarter.run(ForkStarter.java:2
[ERROR]   at org.apache.maven.plugin.surefire.booterclient.ForkStarter.run(ForkStarter.java:2
[ERROR]   at org.apache.maven.plugin.surefire.AbstractSurefireMojo.executeProvider(AbstractSu
[ERROR]   at
org.apache.maven.plugin.surefire.AbstractSurefireMojo.executeAfterPreconditionsChecked(Abstract
[ERROR]   at org.apache.maven.plugin.surefire.AbstractSurefireMojo.execute(AbstractSurefireMo
[ERROR]   at org.apache.maven.plugin.DefaultBuildPluginManager.executeMojo(DefaultBuildPlugin
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.doExecute2(MojoExecutor.java:32
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.doExecute(MojoExecutor.java:316
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.execute(MojoExecutor.java:212)
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.execute(MojoExecutor.java:174)
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.access$000(MojoExecutor.java:75
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor$1.run(MojoExecutor.java:162)
[ERROR]   at org.apache.maven.plugin.DefaultMojoExecutionStrategy.execute(DefaultMojoExecut
[ERROR]   at org.apache.maven.lifecycle.internal.MojoExecutor.execute(MojoExecutor.java:159)
[ERROR]   at
org.apache.maven.lifecycle.internal.LifecycleModuleBuilder.buildProject(LifecycleModuleBuilder.
[ERROR]   at org.apache.maven.lifecycle.internal.LifecycleModuleBuilder.buildProject(Lifecycl
[ERROR]   at
org.apache.maven.lifecycle.internal.builder.singlethreaded.SingleThreadedBuilder.build(SingleTh
[ERROR]   at org.apache.maven.lifecycle.internal.LifecycleStarter.execute(LifecycleStarter.ja
[ERROR]   at org.apache.maven.DefaultMaven.doExecute(DefaultMaven.java:261)
[ERROR]   at org.apache.maven.DefaultMaven.doExecute(DefaultMaven.java:173)
[ERROR]   at org.apache.maven.DefaultMaven.execute(DefaultMaven.java:101)
[ERROR]   at org.apache.maven.cli.MavenCli.execute(MavenCli.java:919)
[ERROR]   at org.apache.maven.cli.MavenCli.doMain(MavenCli.java:285)
[ERROR]   at org.apache.maven.cli.MavenCli.main(MavenCli.java:207)
[ERROR]   at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
[ERROR]   at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAcces
[ERROR]   at
java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl
[ERROR]   at java.base/java.lang.reflect.Method.invoke(Method.java:568)
[ERROR]   at org.codehaus.plexus.classworlds.launcher.Launcher.launchEnhanced(Launcher.java:2
[ERROR]   at org.codehaus.plexus.classworlds.launcher.Launcher.launch(Launcher.java:201)
[ERROR]   at org.codehaus.plexus.classworlds.launcher.Launcher.mainWithExitCode(Launcher.java
[ERROR]   at org.codehaus.plexus.classworlds.launcher.Launcher.main(Launcher.java:314)
[ERROR]
[ERROR] -> [Help 1]
```

```
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoExecutionException
```

**Analyse  
IA**

**\*\*Cause\*\* :**  
L'erreur **"Cannot instantiate class suites.integration.EnvironmentServiceIT"** indique qu'aucun constructeur n'est disponible dans la classe de test, probablement parce qu'elle hérite de `AbstractTestNGSpringContextTests` (n'en fournit pas) et que le build surefire tente d'instancier la classe via reflection sans arguments. Les dépendances `@Autowired` ne sont pas injectées si le constructeur n'est pas explicite ni implicite.

**\*\*Correction\*\* :**  
Ajouter un constructeur par défaut explicite (optionnel mais recommandé avec Spring/TestNG), ou hériter de `AbstractTestNGSpringContextTests` (inutile ici car `@SpringBootTest` suffit), et s'assurer que `@Autowired` est utilisé correctement (ce qui est le cas ici – donc simplement ajouter `public` au constructeur) ou supprimer la classe parente.

**\*\*Conseil\*\* :**  
Éviter l'héritage de `AbstractTestNGSpringContextTests` sauf besoin spécifique (ex: injection de `setApplicationContext`) ; préférer l'approche moderne `@SpringBootTest` + `@Autowired` seul pour l'instanciation et éviter les erreurs de constructeur.

## Corrections automatiques par IA

L'IA a detecte des erreurs dans certains scripts generes et les a corriges automatiquement.

### create project

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
3421550725211656858/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest.java:[79,9]
cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-
3421550725211656858/Backend/ms_gestion/src/test/java/suites/unit/ProjectServiceCreateHappyTest
Résultat: Script corrigé (3412 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

### create envirenemnt

1 correction(s)

FAILURE

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-
12815052988845666633/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[69,29]
incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-
testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-1281
Résultat: Script corrigé (4701 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

L'erreur n'est plus corrigeable automatiquement. Arrêt du retry.

# Metriques

## Top 5 des tests les plus lents

| Test               | Suite             | Duree  |
|--------------------|-------------------|--------|
| create project     | Suite UNIT        | 11.5 s |
| create envirenemnt | Suite INTEGRATION | 9.3 s  |

## Temps d'execution par type

| Type        | Duree moyenne |
|-------------|---------------|
| UNIT        | 11.5 s        |
| INTEGRATION | 9.3 s         |

## Tests instables

Aucun test instable detecte

## Recommandations

- create envirenemnt:

**\*\*Cause\*\*** :

L'erreur `Cannot instantiate class suites.integration.EnvironmentServiceIT` indique qu'aucun constructeur par défaut n'est disponible dans la classe de test, probablement parce qu'elle hérite de `AbstractTestNGSpringContextTests` (qui n'en fournit pas) et que le build surefire tente d'instancier la classe via reflection sans arguments — or, les dépendances `@Autowired` ne sont pas injectées si le constructeur n'est pas explicite ni implicite.

**\*\*Correction\*\*** :

Ajouter un constructeur par défaut explicite (optionnel mais recommandé avec Spring/TestNG), ou mieux : supprimer l'héritage de `AbstractTestNGSpringContextTests` (inutile ici car `@SpringBootTest` suffit), et s'assurer que `@Autowired` est utilisé correctement (ce qui est le cas ici — donc simplement ajouter `public EnvironmentServiceIT()` ou supprimer la classe parente).

**\*\*Conseil\*\*** :

Éviter l'héritage de `AbstractTestNGSpringContextTests` sauf besoin spécifique (ex: injection de contexte via `setApplicationContext`) ; préférer l'approche moderne `@SpringBootTest` + `@Autowired` seul pour simplifier l'instanciation et éviter les erreurs de constructeur.

## Annexe - Scripts generes par IA

### create project

```
package suites.unit;

import com.pfe.platform.ms_gestion.dto.request.CreateProjectRequest;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.security.SecurityUtils;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;
import com.pfe.platform.ms_gestion.service.ProjectService;

import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.MockedStatic;
import org.mockito.MockitoAnnotations;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Optional;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.Mockito.*;
import static org.testng.Assert.*;

public class ProjectServiceCreateHappyTest {

    @Mock
    private ProjectRepository projectRepository;

    @Mock
    private ProjectMemberRepository projectMemberRepository;

    @Mock
    private ProjectAccessService projectAccessService;

    @InjectMocks
    private ProjectService projectService;

    private AutoCloseable mocks;
    private MockedStatic<SecurityUtils> mockedSecurity;

    @BeforeMethod
    public void setUp() {
        mocks = MockitoAnnotations.openMocks(this);
        mockedSecurity = mockStatic(SecurityUtils.class);
        mockedSecurity.when(SecurityUtils::getCurrentUserId).thenReturn(1L);
    }

    @AfterMethod
    public void tearDown() throws Exception {
        mockedSecurity.close();
        mocks.close();
    }

    @Test
    public void test_create_happy_path() {
        // Given
        CreateProjectRequest request = new CreateProjectRequest();
        request.setName("Projet Alpha");
        request.setDescription("Développement d'une nouvelle application web");

        when(projectRepository.existsByName("Projet Alpha")).thenReturn(false);
```

```

Project savedProject = new Project();
savedProject.setId(1L);
savedProject.setName("Projet Alpha");
savedProject.setDescription("Développement d'une nouvelle application web");
savedProject.setCreatedBy(1L);
when(projectRepository.save(any(Project.class))).thenReturn(savedProject);

ProjectMember savedMember = new ProjectMember();
savedMember.setId(1L);
savedMember.setProject(savedProject);
savedMember.setUserId(1L);
when(projectMemberRepository.save(any(ProjectMember.class))).thenReturn(savedMember);

// When
ProjectResponse result = projectService.create(request);

// Then
assertNotNull(result);
assertEquals(result.getName(), "Projet Alpha");
assertEquals(result.getDescription(), "Développement d'une nouvelle application web");
assertEquals(result.getCreatedBy(), Long.valueOf(1L));
verify(projectRepository).existsByName("Projet Alpha");
verify(projectRepository).save(any(Project.class));
verify(projectMemberRepository).save(any(ProjectMember.class));
}
}

```

## create environemnt

```

package suites.integration;

import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.entity.ProjectMemberId;
import com.pfe.platform.ms_gestion.repository.CampaignRepository;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextHolderImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.TestPropertySource;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest
@ActiveProfiles("test")
@Transactional
@Rollback(true)
@TestPropertySource(properties = {
    "spring.datasource.url=jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1;MODE=PostgreSQL",
    "spring.datasource.driverClassName=org.h2.Driver",

```

```
"spring.datasource.username=sa",
"spring.datasource.password=",
"spring.jpa.hibernate.ddl-auto=create-drop",
"spring.jpa.database-platform=org.hibernate.dialect.H2Dialect"
})
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

    @Autowired private EnvironmentService environmentService;
    @Autowired private ProjectRepository projectRepository;
    @Autowired private ProjectMemberRepository projectMemberRepository;
    @Autowired private EnvironmentRepository environmentRepository;
    @Autowired private CampaignRepository campaignRepository;

    private Long projectId;
    private Long userId = 1L;

    @BeforeMethod
    public void setUp() {
        // Insert prerequisite Project
        Project project = new Project();
        project.setName("AddProject");
        project.setDescription("Test project");
        project.setGitRepoUrl("https://github.com/org/webapp");
        project.setCreatedBy(userId);
        project = projectRepository.save(project);
        projectId = project.getId();

        // Insert ProjectMember to satisfy checkMembership
        ProjectMember projectMember = new ProjectMember();
        projectMember.setId(new ProjectMemberId(userId, projectId));
        projectMember.setProject(project);
        projectMemberRepository.save(projectMember);

        // Set security context
        SecurityContextHolder.setContext(new SecurityContextImpl(
            new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
    }

    @AfterMethod
    public void tearDown() {
        SecurityContextHolder.clearContext();
    }

    @Test
    public void test_add_happy_path() {
        // Given
        CreateEnvironmentRequest request = new CreateEnvironmentRequest();
        request.setName("AddProject");
        request.setBaseUrlWeb("https://app.example.com");
        request.setBaseUrlApi("https://api.example.com");
        request.setGitRepoUrl("https://github.com/org/webapp");
        request.setGitBranch("develop");
        request.setDatabaseType("POSTGRESQL");

        // When
        EnvironmentResponse result = environmentService.add(projectId, request);

        // Then
        assertNotNull(result);
        assertNotNull(result.getId());
        assertEquals(result.getName(), "AddProject");
        assertEquals(result.getBaseUrlWeb(), "https://app.example.com");
        assertEquals(result.getBaseUrlApi(), "https://api.example.com");
        assertEquals(result.getGitRepoUrl(), "https://github.com/org/webapp");
        assertEquals(result.getGitBranch(), "develop");
        assertEquals(result.getDatabaseType(), "POSTGRESQL");

        // Verify persistence in DB
        Environment savedEnv = environmentRepository.findById(result.getId()).orElse(null);
        assertNotNull(savedEnv);
    }
}
```

```
assertEquals(savedEnv.getName(), "AddProject");
assertEquals(savedEnv.getProject().getId(), projectId);
}
}
```